

Limits – 7

Means, inequalities and contractions

Homework

1 сентября 2026 года

Theoretical minimum. Power means and the inequalities between them; the Cauchy–Schwarz, Jensen and Chebyshev sum inequalities; Toeplitz’s and Kronecker’s lemmas; the contraction principle and its weakenings; the Cauchy criterion as a way of proving convergence without knowing the limit.

1. Let $a_n > 0$ with $\frac{a_1 + \dots + a_n}{n} \rightarrow a$. Is it true that $\sqrt[n]{a_1 \cdots a_n} \rightarrow a$? Prove it or construct a counterexample.
2. Prove Carleman’s inequality $\sum_{n \geq 1} \sqrt[n]{a_1 \cdots a_n} \leq e \sum_{n \geq 1} a_n$ for non-negative a_n , and show that the constant e cannot be improved.
3. Let $x_1 = 1$ and $x_{n+1} = \frac{x_n + 2/x_n}{2}$. Prove that $|x_n - \sqrt{2}| \leq \frac{1}{2^{2^n}}$ from some index on.
4. Prove Chebyshev’s sum inequality: if $a_1 \leq \dots \leq a_n$ and $b_1 \leq \dots \leq b_n$, then $n \sum a_i b_i \geq (\sum a_i)(\sum b_i)$; deduce the inequality between M_1 and M_2 .
5. Let $x_n > 0$ satisfy $x_{n+1} \leq \frac{x_n + x_{n-1}}{2}$. Prove that the sequence converges and express its limit through x_0 and x_1 .
6. Let $a_n \rightarrow a$ and $b_n \rightarrow b$. Prove that $\frac{1}{n} \sum_{k=1}^n a_k b_{n-k+1} \rightarrow ab$ and explain why this is a special case of Toeplitz’s lemma.
7. Prove Jensen’s inequality for a convex function and a sequence of weights, and deduce from it Hölder’s inequality and the inequality between means as special cases.

8. Let $f: [0, 1] \rightarrow [0, 1]$ satisfy $|f(x) - f(y)| < |x - y|$ for $x \neq y$. Prove that the iterates converge to the unique fixed point although f is not a contraction.
9. Let $\sum a_n$ be a convergent series with positive terms. Prove that $\sum \sqrt{a_n a_{n+1}}$ converges, and that $\sum \sqrt{a_n}/n$ converges by Cauchy–Schwarz; give an example in which $\sum \sqrt{a_n}$ diverges.
10. Let $x_n \in [0, 1]$ satisfy $x_{n+1} \geq \frac{x_n + x_{n+2}}{2}$ for all n (discrete concavity). Prove that x_n converges and that $n(x_n - x_{n+1}) \rightarrow 0$.